

LUND UNIVERSITY

Mathematics, Faculty of Science

**Number theory**
**MATM35**
**HT 2021**
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# Written examination
# January 11, 2022

1. Determine, using Legendre symbols, whether the congruences

$$x^2 + 4x - 11 \equiv 0 \pmod{577} \quad \text{and} \quad x^2 + 4x - 11 \equiv 0 \pmod{599},$$

respectively, are solvable or not. (Note that 577 and 599 are primes.)

2. Determine the infinite continued fraction representation of the number $\dfrac{\sqrt{35}}{5}$.

3. Let $p$ be a prime of the form $p = 2^n + 1$ for a positive integer $n$. Prove that every integer $a$ with $\gcd(a, p) = 1$ is either a quadratic residue or a primitive root (mod $p$).

4. Let $p$ be an odd prime and let $r_1, r_2$ be primitive roots (mod $p$). Show that their product $r_1 r_2$ is not a primitive root (mod $p$).

5. (a) Prove that every positive integer $n$ can be expressed as $n = a^2 + b^2 + c^2 + 4d^2$, where $a, b, c, d \in \mathbb{Z}$.

   (b) Find and prove a criterion for a positive integer $n$ to be expressible as $n = a^2 + 4b^2$, where $a, b \in \mathbb{Z}$.

6. The Dedekind $\psi$-function is defined by $\psi(n) = n \prod_{\substack{p \mid n \\ p \text{ prime}}} \left(1 + \dfrac{1}{p}\right)$.

   (a) Prove that

$$\psi(n) \leq \sigma(n)$$

   for all $n$, with equality if and only if $n$ is squarefree. (Here, $\sigma(n) = \sum_{d \mid n} d$ is the sum-of-divisors function.)

   (b) Prove that

$$2\psi(n) \geq n + \sigma(n)$$

   for all $n$.

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Give careful motivations for your solutions!